MIL-H-21722A(AS)

25 April 1969
Superseding
MIL-H-21722(AER)
17 November 1958

MILITARY SPECIFICATION

HOOKS, THRUST AND STABILIZING, JATO-4500 POUNDS NOMINAL THRUST

This Specification has been approved by
the Naval Air Systems Command,
Department of the Navy.

1. SCOPE

1.1 This Specification covers the manufacture, inspection and testing requirements for hooks, thrust and stabilizing, JATO-4500 pounds nominal thrust suspension system.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-378	Plastic Film (Polyethylene Thin Gage)
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-C-843	Cushioning Material, Cellulosic

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000/1	Drawing, Engineering and Associated Data

FSC 1680

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SPECIFICATIONS

Military - (Cont'd)

MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible

STANDARDS

Federal

FED-STD-151	Metals, Test Methods

Military

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of

DRAWINGS

NAVAIR 67A301D1	Fixture, Test, Vibrations, Life, Release Function
NAVAIR 67A302D2	Fixture, Test, Static, Repeat Load
NAF 603894	Hook, Thrust, Jato, Retractable
NAF 603922	Hook, Stabilizing, Jato, Retractable

(Copies of Specifications, Standards, Drawings, and Publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First Article - The hooks furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein.

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3.1.1 Manufacturing - The thrust hooks shall be manufactured in accordance with the latest revision of drawings NAF 603894. The stabilizer hooks shall be manufactured in accordance with the latest revision of drawing NAF 603922.

3.1.2 Selection of specifications and standards - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2 Materials - Materials shall conform to the applicable specifications, shall be as specified herein, and on the applicable drawings. Materials which are not covered by applicable specifications or standards shall be of the best commercial quality, the lightest practical weight, and entirely suitable for the purpose intended.

3.3.1 AN or MIL Standard Parts - AN or MIL Standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etcetera, may be used, provided they possess suitable properties and are replaceable by the AN or MIL Standard Parts without alteration, and provided the corresponding AN or MIL part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding AN or MIL Standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Interchangeability of Parts - All parts having the same manufacturer's part number shall be functionally and physically interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of MIL-D-1000/1.

3.4 Performance - The Jato hooks shall meet the performance requirements specified in Section 4 when subjected to the applicable tests.

3.5 Weight - The weight of the thrust hooks shall not exceed 4.0 pounds. The weight of the stabilizer hooks shall not exceed 1.6 pounds.

3.6 Identification - Each hook shall be identified on the frame of the hook with the contract number and the contractor's code number. A method of marking shall be selected which will not affect the structural integrity or the performance of the hook and which cannot easily be removed.

3.7 Lubrication - Lubricant Molykote G manufactured by the Alpha-Molykote Corporation, Stamford, Connecticut, or equivalent, shall be used. The hooks shall be lubricated as follows:

(a) Prior to assembly of the thrust hook apply a thin coat of the lubricant to the working surfaces of parts

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603894-12 (keeper), 603894-11 (lever), 603894-20 (hook-plate), and 603894-3 (latch assembly). Also apply a thin coat of lubricant to the diameter of pins 603894-29, 603894-30, and 603894-31. After assembly, actuate the hook several times. This will spread the lubricant evenly and work it into the surfaces of the parts.

(b) Prior to assembly of the stabilizing hooks apply a thin coat of the lubricant on the work surfaces of 603922-3 (assembly), 603922-10 (sear), and 603922-4 (hook). Also apply a thin coat of lubricant on the diameter of pins 603922-7 and 603922-17. After assembly, actuate the hook several times. This will spread the lubricant evenly and work it into the surfaces of the parts.

(c) After assembly there will be no small gobs of lubricant clinging to the sides of the parts.

(d) The use of the term working surfaces, as used herein, defines those surfaces of levers or parts which come into contact with other moving parts of the assembly.

3.8 Test Fixtures -

3.8.1 Life Test - The life test fixture and the hydraulic jettison system used in conducting the life test (4.6.4) shall be in accordance with NAVAIR DWG 67A301D1. The test fixture shall be mounted on a level support capable of absorbing the jettison forces.

3.8.2 Vibration Test - The test fixture used to perform the vibration test (4.6.3) shall be the same as that used for the life test. The vibration input monitoring location shall be as indicated in NAVAIR DWG 67A301D1.

3.8.3 Static and repeat load test - The test fixture used to perform the static and repeat load test (4.6.5) shall be constructed in accordance with NAVAIR DWG 67A302D2.

3.9 Workmanship - Any variation in uniformity of hooks, sharp edges, burrs, surface pits, inferior plating or any other visible evidence of careless workmanship shall be cause for rejection.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the Assigned Test Activity (6.3.1) is responsible

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for the performance of all inspection requirements as specified herein. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection terms - Military Standard MIL-STD-109 shall govern the definition of terms used herein.

4.2 Classification of inspections - The inspection of JATO hooks shall be classified as follows:

(a) First Article Inspection

(b) Quality conformance inspection

4.2.1 First Article Inspection - First Article Inspections are those inspections performed after award of contract on sample hooks to determine that the hooks meet the requirements of this specification.

4.2.2 Quality conformance inspection - Quality conformance inspections are those inspections performed on hooks manufactured and submitted for acceptance under contract.

4.3 Inspection of Materials -

4.3.1 Commercial Materials - When materials are procured under commercially recognized non-governmental specifications, such as SAE numbers a certificate showing compliance with such specifications shall be furnished by the supplier of the material. Such a certificate shall be in a form acceptable to the Government Inspector.

4.4 First Article Inspection - First article inspections shall consist of the inspections described in 4.6 in the order listed in Table I. The hooks shall be plainly identified by securely attached durable tags marked with the following information:

Sample submitted by (Name) (Date) for first article inspection in accordance with the requirements of MIL-H-21722A(AS) and Dwg NAF under contract No.

4.4.1 First Article sampling - The first three thrust hooks and the first six stabilizer hooks produced under each contract and fabricated from production tools shall be first article test samples. The samples shall be identical,

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to the product to be supplied, under contract or order, in design, construction, quality material, workmanship, and method of manufacture. Deviation from the standards of the first article sample shall be permitted only by the procuring activity. Evidence of unauthorized change shall be cause for rejection. These samples shall be delivered, for inspection and testing, to the assigned test activity (6.3.1) with certification by the cognizant inspector that the materials used in their manufacture are as specified on applicable drawings. Hooks shall be selected for testing in sets consisting of one thrust and two stabilizer hooks. Samples subjected to first article tests shall not be considered or shipped as part of the contract or order.

TABLE I

TITLE	APPLICABLE PARAGRAPH	SET 1	SET 2	SET 3
Physical Characteristics (Assembled)	4.6.1	X	X	X
Vibration	4.6.3	X		
Life	4.6.4		X	X
Salt Spray	4.6.2		X	
Static and Repeat Load	4.6.5		X	X
Destruction, Static and Repeat Load	4.6.6		X	X
Physical Characteristics (Disassembled)	4.6.1.1	X		

4.4.2 Correction of defects - Any defects discovered during the inspection and testing of the First Article Test samples shall be corrected by the contractor. Such measures as may be found necessary to correct these defects shall likewise be incorporated in all subsequent hooks offered for acceptance under contract.

4.5 Quality conformance inspection - Quality conformance inspections shall consist of:

(a) Examination of Product (4.5.2)

(b) Sampling inspection (4.5.1)

(c) Magnetic particle inspection (4.7)

4.5.1 Sampling inspection - Samples shall be selected by the Government Inspector at random, from each production lot, in accordance with Table II. Samples shall be forwarded to the assigned test activity (6.3.1) and tested in the order listed in Table III.

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TABLE II

TOTAL NO. OF HOOKS	NO. OF SAMPLES SELECTED	
	THRUST	STABILIZER
99 or less	2	4
100 to 299	3	6
300 to 500	4	8

Note: When the lot size exceeds 500 hooks, the lot shall be subdivided into lots of 500 units and the selection for each subdivision shall be in accordance with the above table.

TABLE III

TITLE	APPLICABLE PARAGRAPH	SET 1	SET 2	SET 3	SET 4
Physical Characteristics (Assembled)	4.6.1	X	X	X	X
Life Test	4.6.4		X	X	
Static and Repeat Load	4.6.5	X		X	
Destruction	4.6.6		X		X
Physical Characteristics (Disassembled)	4.6.1.1	X			

4.5.2 Examination of product - Each hook submitted for acceptance under contract shall be inspected visually, by the contractor, to insure that it has been properly assembled and that the workmanship and the finishes are in accordance with the provisions of this specification. Samples subjected to quality conformance tests shall not be considered or shipped as part of the contract or order.

4.6 Inspection methods -

4.6.1 Physical characteristics - The units submitted for this test shall be photographed to illustrate clearly appearance and general arrangement. The test samples shall be weighed and subjected to an envelope and mounting hole dimensional check. In addition each assembled test sample shall be visually examined to insure it has been lubricated, properly assembled, and the workmanship and finishes are in accordance with the applicable drawings. Sufficient care must be taken to prevent disturbance of the lubricant.

4.6.1.1 At the completion of all first article and quality conformance tests, one each of the thrust and stabilizing hooks shall be disassembled and the parts inspected and tested to insure that surface finish, plating and hardness are in accordance with applicable drawings.

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4.6.2 Salt spray test - A salt spray test shall be made on Set No. 2 in accordance with Federal Test Method Standard 151a, Method 811.1, after completion of 400 cycles of the life test (4.6.4). The length of the salt spray test shall be 50 hours. At the end of the test period the hooks shall be inspected, photographed and manually operated. The performance of the hook during manual operation shall be noted. Salt deposits resulting from the exposure conditions shall be removed by rinsing with tapwater. Without disassembling, the working surfaces (3.7.d), of the hooks shall then be lubricated and the life test continued to 500 cycles.

4.6.3 Vibration test - A vibration test shall be performed on the JATO hooks using test fixture shown on NAVAIR Dwg 67A301D1. An expended 225 pound JATO bottle or a 225 pound mock JATO bottle shall be installed on the test fixture. The JATO hooks shall be vibrated for a period of twenty-five (25) hours as follows:

(a) Vibration input frequency, 33 $^{+0}_{-5}$ CPS

(b) Vibration input amplitude, .032 inches at 33 CPS

(c) Direction of vibration shall be vertical

The vibration input amplitude shall be measured at the point of inherent rigidity as indicated in NAVAIRDWG 67A301D1. Should undue resonance generated by either the test set-up or the JATO bottle occur at 33 CPS, reduce the input frequency to eliminate the resonance and adjust the displacement to produce the same G level. The test duration shall then be adjusted to produce an equivalent number of cycles of vibration based on 33 CPS for 25 hours. Fracture of hook components or release of bottle during test period due to vibration, mechanical failure, or loosening of parts shall be considered cause for rejection of test hooks. Prior to and upon completion of the vibration test the hooks shall be manually actuated three times to demonstrate satisfactory operation.

4.6.4 Life test - The life test shall be conducted at room temperature on the life test fixture shown on NAVAIR Dwg 67A301D1. The test shall be conducted as follows:

(a) Attach a 225 pound expended JATO bottle, or a mock bottle of equal weight, to the JATO hooks.

(b) Release JATO bottle

(c) Repeat (a) and (b) for a total of 500 operations.

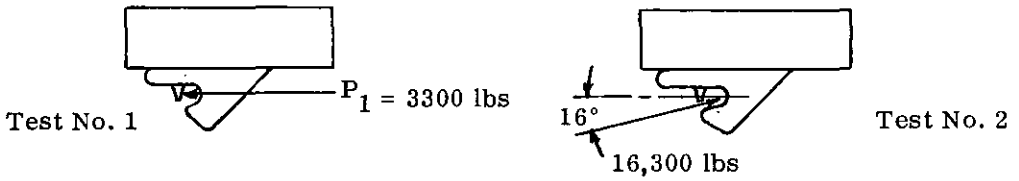
4.6.4.1 The force required to release the JATO bottle shall not exceed the force obtained with the hydraulic jettison system illustrated on NAVAIR Dwg 67A301D1. The time required to jettison the bottle shall be in the range of 70 - 150 milliseconds.

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4.6.5 Static and Repeat Load Tests -

4.6.5.1 Static load tests shall be conducted at room temperature on the test fixture shown on NAVAIR Dwg 67A302D2. The test shall be conducted as follows:

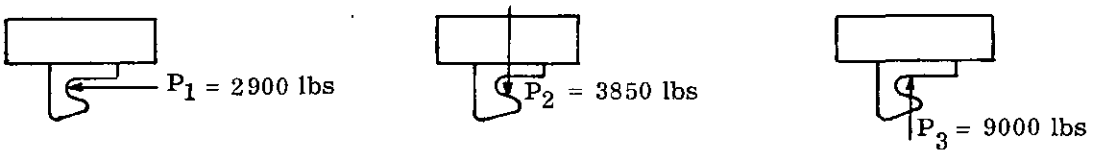
4.6.5.1.1 Thrust hook (NAF 603894) - Loads shall be applied as shown below:



Test No. 1 Test No. 2

When subjected to the above tests the thrust hook shall show no evidence of mechanical or material failure.

4.6.5.1.2 Stabilizing hook (NAF 603922) - Loads shall be applied as shown below:



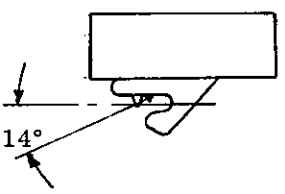
Test No. 1 Test No. 2 Test No. 3

When subjected to the above tests the stabilizing hook shall show no evidence of mechanical or material failure.

4.6.5.1.3 Upon completion of the static load tests the hooks shall be manually actuated three times to demonstrate satisfactory operation.

4.6.5.2 Repeat load tests shall be conducted at room temperature on the test fixture shown on NAVIAR Dwg 67A302D2. The tests shall be conducted as follows:

4.6.5.2.1 Thrust hook (NAF 603894) - Loads shall be applied as shown below:



P = 10,750 lbs

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The repeat load tests are cycled from zero to the specified load and back to zero. There shall be no evidence of mechanical or material failure below 2000 cycles.

Upon completion of the repeat load test the hooks shall be manually actuated three times to demonstrate satisfactory operation.

4.6.5.2.2 Stabilizing hook (NAF 603922) - Loads shall be applied as shown below:



Condition	P_1	P_2	P_3
I	---	2100 lb	1200 lb
II	1700 lb	---	1200 lb

The repeat load tests are cycled from zero to the specified load and back to zero. There shall be no evidence of mechanical or material failure below 2000 cycles for Condition I and Condition II.

Upon completion of the repeat load tests the hooks shall be manually actuated three times to demonstrate satisfactory operation.

4.6.6 Destruction tests -

4.6.6.1 Static destruction tests -

(a) Thrust Hook (NAF 603894) - Loads shall be applied in the same manner as indicated in 4.6.5.1.1. Failure to restrain the load shall not occur below values of P_1 = 4000 lbs and P_2 = 17,000 lbs.

(b) Stabilizing hook (NAF 603922) - Loads shall be applied in the same manner as indicated in 4.6.5.1.2. Failure to restrain the load shall not occur below values of P_2 = 5200 lbs, and P_3 = 9450 lbs.

4.6.6.2 Repeat load destruction tests -

(a) Thrust hook (NAF 603894) - Magnitude and direction of the load to be as shown in 4.6.5.2.1. Failure to restrain the load shall not occur below 5,000 cycles for first article testing, 2,500 cycles for quality conformance testing but to be continued to 5,000 cycles or failure, whichever occurs first, for reliability purposes.

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(b) Stabilizing hook (NAF 603922) - Magnitude and direction of loads to be shown in 4.6.5.2.2. Failure to restrain the load shall not occur below 5,000 cycles in Condition I for preproduction testing, 2,500 cycles for quality conformance testing but to be continued to 5,000 cycles or failure, whichever occurs first, for reliability purposes.

4.7 Magnetic Particle Inspection - After machining and prior to plating, the contractor shall perform magnetic particle inspection on parts as indicated on applicable drawings. The contractor shall make the inspection certificates available to the Government Inspector.

4.8 Rejection and Retest -

4.8.1 Quality conformance test samples - The failure of any sample hook to pass the quality conformance inspections and tests outlined in 4.5 shall be cause for rejection of all hooks represented by that sample, subject to retests as provided in 4.8.2.

4.8.2 Retest - In the event any sample representing a production lot fails to pass the quality conformance inspection and tests outlined in 4.4 the contractor shall have the opportunity of reworking the represented lot and resubmitting to test twice as many samples, selected at random by the Government Inspector, as originally submitted for the lot. Should all of these samples pass the required tests, the lot represented thereby shall be considered acceptable. Should any of these samples fail, the entire lot represented thereby shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - Preservation and packaging shall be Level A or C as specified by the procuring activity (see 6.2).

5.1.1 Level A - Each thrust or stabilizing hook shall be individually packaged in accordance with MIL-P-116, Method III. The hook shall be placed within a transparent film bag of .0015 inch minimum gauge and heat sealed. Bag material shall conform to L-P-378 or the commercial equivalent. Each bagged hook shall be adequately cushioned, material conforming to PPP-C-843, and packaged within an individual close fitting fiberboard box conforming to PPP-B-636, Style OPF, Class Domestic, Grade 125; closure shall be accomplished in accordance with applicable procedures in the appendix.

5.1.2 Level C - Each hook shall be preserved and packaged in a manner that will prevent deterioration and damage during transit from the supply source to the first receiving activity. The contractor's commercial practice may be utilized when it conforms to the criteria stated herein.

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5.2 Packing - Packing shall be Level A, B or C as specified by the procuring activity (see 6.2). Insofar as practicable, shipping containers shall effect a close fit, contain identical quantities and be of uniform dimensional configuration.

5.2.1 Level A - Hooks packaged as specified in 5.1.1 shall be packed for shipment in fiberboard containers conforming to PPP-B-636, Class - Weather resistant, Grade V3s/3Vc or PPP-B-640, Class 2, Grade A. Each container shall be fitted with a waterproof liner conforming to MIL-L-10547. The gross weight of each packed container shall not exceed 70 pounds for the PPP-B-636 container nor 150 pounds for the PPP-B-640 container. The closure, sealing and reinforcing (banding) of the containers shall be accomplished in accordance with the procedure specified in the appendix of the applicable container specification.

5.2.2 Level B - Hooks packaged as specified in 5.1.1 shall be packed for shipment in fiberboard containers conforming to PPP-B-636, Class - Domestic, Grade 275 or PPP-B-640, Class 1, Grade A or B. The gross weight of each packed container shall not exceed 90 pounds for the PPP-B-636 containers nor 200 pounds for PPP-B-640 containers. The closure and reinforcing (banding) of the containers shall be accomplished in accordance with the procedure specified in the appendix of the applicable container specification.

5.2.3 Level C - Hooks packaged as specified shall be packed for shipment in a manner that will afford protection against damage during transit from the supply source to the first receiving activity. The pack shall, as a minimum, comply with the rules and regulations applicable to the mode of transportation utilized.

5.3 Marking - Each unit and shipping container shall be marked in accordance with MIL-STD-129. In addition, the following marking shall be required on the unit packages:

Specification MIL-H-21722A(AS)
Contract/Order No.
FSN 1680-697-3216-x110-Thrust Hook
FSN 1680-697-3217-x110-Stabilizing Hook
{as applicable}

6. NOTES

6.1 Intended use - The stabilizing and thrust hooks covered by this specification are intended for use on certain aircraft to attach the Jet Assisted Take-Off (JATO) bottles.

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6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Testing activity, delivery of test reports. (See 4.1)
(c) Level of packaging and packing required (See 5.1 and 5.2)
(d) Delivery information concerning test samples.
(e) Hooks subjected to first article or quality conformance tests shall not be considered or shipped as part of the contract or order.

6.3 Definitions -

6.3.1 Assigned Test Activity - The assigned test activity is the activity designated by the procuring activity to conduct the first article, quality conformance and sampling inspections listed herein.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION MIL-H-21722A(AS)

HOOKS, THRUST AND STABILIZING, JATO-4500 POUNDS NOMINAL THRUST

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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